Sayantan Bhattacharya

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EDUCATION

Techno Main Salt Lake — Kolkata, India

B.Tech in Computer Science and Engineering (AI and Machine Learning) — *CGPA: 7.13 / 10*

Expected June 2027

- Relevant coursework: Machine Learning, Deep Learning, NLP, Neural Networks, Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Database Management Systems, Web Development, Computer Networks, Operating Systems

Test Scores: GRE : 304/340 - 156 Q ,148 V ,4.0 A.W.A — August 27, 2026

TOEFL : 100/120 - 5.0/6.0 overall - CEFR C1 - 5/6 R ,4.5/6 L ,5.5/6 W , 5/6 S — October 03 2026

EXPERIENCE

Independent Technical Contractor — 2022 — 2026

Self-Employed — Video Editing, India

- Automated a multi-stage post-production pipeline using scripted DaVinci Resolve workflows, reducing per project processing time by 40% across 50+ client deliverables — applied systematic optimization and iterative testing to identify and eliminate bottlenecks in a complex, multi-input system.
- Managed end-to-end delivery of 30+ long-form projects across international clients by building a structured scoping and revision framework, achieving zero late deliveries over 4 years.
- Designed reusable modular templates for color grading, motion, and audio processing by abstracting recurring patterns into parameterized workflows, scaling output 3x without proportional time increase.

PROJECTS

Covid CT Audit — Auditing a 99.76%-Accurate COVID-19 CT Classifier — 2026

Python, PyTorch , MobileNetV3, scikit-learn, Grad-CAM, pandas — *Personal Project*

- Showed that slice-level splits inflate accuracy by placing near-duplicate slices in both train and test: recovered pseudo-patient groups by clustering deep and thumbnail features, and measured accuracy falling from 99.42% to 95.39% under grouped splits (error up to 8.0x).
- Identified shortcut learning: a metadata-only model reached 0.996 balanced accuracy on random splits (0.630 under grouped splits), and the classifier kept ~0.93 balanced accuracy with the lungs masked out, confirmed with Grad-CAM lung-enrichment analysis.
- Tested external validity and calibration on Sao Paulo and Tehran (COVID-CT-MD) data: AUROC ≈0.83 externally, near chance (0.43-0.49) for COVID vs community-acquired pneumonia, and calibration error rising from 0.02 to 0.19-0.53; released a one-script reproducible pipeline and a 7-step audit protocol.

Github: github.com/PopcornKitsilano/covid-ct-audit ; Research Paper Publication

PassME — Full-Stack Password Manager — 2025

Python, Flask, SQLite, HTML, CSS, JavaScript — *CS50x Final Project*

- Achieved zero critical vulnerabilities across all OWASP Top 10 attack vectors by implementing bcrypt hashing, Fernet encryption, and parameterized queries throughout the authentication and storage pipeline.
- Reduced credential retrieval time to under 100ms by designing an indexed SQLite schema with optimized CRUD operations, supporting password generation, copy-to-clipboard, and session management.
- Delivered a fully functional, security-compliant password manager as a solo developer, verified through manual penetration testing of all input surfaces.

Youtube Demo : youtube.com/watch?v=X2SUDquMxqc

Semantic Search RAG Chatbot — Two-Stage Retrieval Experiment 2026

Python, Flask, ChromaDB, LangChain, Groq (LLaMA-3.3 70B), Scrapy, HuggingFace Transformers *Personal Project*

- Improved retrieval precision over single-stage cosine similarity search by designing a two-stage pipeline — ChromaDB vector search over 50 candidates followed by CrossEncoder reranking (ms-marco-MiniLM-L-6-v2) — with a Scrapy and Playwright crawler using SHA-256 hashing to avoid redundant re-indexing.
- Enabled coherent multi-turn interaction via a conversation memory module preserving the last 3 Q&A pairs, generating cited answers with LLaMA-3.3 70B via Groq API across PDF, URL, and crawled web sources.

GitHub: github.com/PopcornKitsilano/rag-chatbot

BasketBawl — Basketball Hooping Game 2024

MIT Scratch *CS50x PSET0*

- Built a real-time score-tracking game with keyboard-driven player movement (A/D) and shooting (1), 2 points per successful shot.
- Implemented conditional game flow with name-based prompts and graceful exit handling for non-affirmative user responses.

Link: scratch.mit.edu/projects/1061387522

Razer Crimson Royale — Custom Hardware Lighting Engine 2026

Python, Razer Chroma REST API *Personal Project*

- Built a real-time per-key animation engine delivering 20 effects at 7fps by implementing HSV color math, directional transforms, and a session-based REST protocol — demonstrating applied systems programming outside coursework.
- Eliminated Razer Synapse dependency by reverse-engineering the local Chroma SDK REST API, implementing BGR encoding and a terminal-driven playlist system with full runtime control over effect order, speed, and direction.

GitHub: github.com/PopcornKitsilano/razer-crimson-royale

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, Java, SQL, Bash

AI and ML: Machine Learning, Deep Learning, NLP, Computer Vision, Neural Networks, LLM frameworks, LangChain, Prompt Engineering, Model development and deployment, AWS SageMaker, AWS Bedrock, Conversational AI, Generative AI, Supervised and Unsupervised Learning, Feature Engineering

ML Libraries: NumPy, pandas, scikit-learn, Matplotlib, TensorFlow, PyTorch, Hugging Face Transformers

Web and Frameworks: Flask, Django, React, Node.js, HTML5, CSS3, Bootstrap, REST APIs

Databases: SQLite, SQL (schema design, indexing, joins, CRUD operations)

Cybersecurity: Penetration testing, OWASP Top 10, vulnerability assessment, SIEM, log analysis

Security Tools: BurpSuite, Metasploit, Wireshark, Hydra, CAPA, Splunk, Graylog

Developer Tools: Git, GitHub, Linux (Arch, Kali), DaVinci Resolve, Arduino, Agile, VS Code

Concepts: OOP, Data Structures and Algorithms, Software Design, Secure Coding

CERTIFICATIONS

CS50X — Introduction to Computer Science Aug 2025

Harvard University via edX | View Certificate | Verified PDF

Getting Started with Artificial Intelligence Nov 2024

IBM via Credly | View Certificate

PreSecurity Certificate July 2025

TryHackMe — 100+ rooms; real-world attack simulation, privilege escalation | View Certificate

HOBBIES

Making YouTube Videos | Playing my BASS | Streaming Video Games | Music Sampling | Basketball